

Combination of Access Control and De-Identification for Privacy Preserving in Big Data

Amine Rahmani, GeCoDe Laboratory, Department of Computer Sciences, Dr. Tahar Moulay University of Saida, Algeria

Abdelmalek Amine, GeCoDe Laboratory, Department of Computer Sciences, Dr. Tahar Moulay University of Saida, Algeria

Reda Mohamed Hamou, GeCoDe Laboratory, Department of Computer Sciences, Dr. Tahar Moulay University of Saida, Algeria

ABSTRACT

Despite of its emergence and advantages in various domains, big data still suffers from major disadvantages. Timeless, scalability, and privacy are the main problems that hinder the advance of big data. Privacy preserving has become a wide search era within the scientific community. This paper covers the problem of privacy preserving over big data by combining both access control and data de-identification techniques in order to provide a powerful system. The aim of this system is to carry on all big data properties (volume, variety, velocity, veracity, and value) to ensure protection of users' identities. After many experiments and tests, our system shows high efficiency on detecting and hiding personal information while maintaining the utility of useful data. The remainder of this report is addressed in the presentation of some known works over a privacy preserving domain, the introduction of some basic concepts that are used to build our approach, the presentation of our system, and finally the display and discussion of the main results of our experiments.

KEYWORDS

Access Control, Big Data, Bio-Inspired, Data Perturbation, De-Identification, Privacy Preserving

INTRODUCTION

Today, we are awash in a mounting tide of data. A few years ago, decisions were made by guesswork or painstakingly constructed models of reality, but nowadays this can be traced based on the data itself. Data-Driven Decision-Making is broadly recognized era over the planet and this is ascribable to the growing enthusiasm of the term “Big Data”. Big data is a general term that refers to a set of techniques which have the ability of treating huge volumes of continuously varying heterogeneous data in quasi-real time. Thence, this growing technology knows everything around us. It offers a frighteningly complete picture of us. What we do, where we live, where we go, what we say, and even what we feel and believe in.

Observed with the development of technology and communication means, the amount of cyber-attacks had been doubled by the hundreds in term of type and frequency. Many solutions had been acquired in order to neutralize these attacks. Yet, these solutions are being rapidly rendered obsolete against attacks over big data services. For that, many researches have been adopted on focusing on the security in big data. Privacy preserving is now a broadly known research area. The intent of this area is to evolve techniques for rigid control in order to keep large data from becoming the “big brother”.

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Privacy preserving refers to a lot of techniques and algorithms that consist of hiding sensitive data within a shared data either by controlling the access to it, or using data perturbation methods.

A specific area of data perturbation is known by de-identification. The de-identification is a word that refers to the fact of removing or hiding data in order to prevent any disclosure of identifiable data. This concept can be seen as a special topic for researches where the goal is to find an effective approach that ensures the maximum of de-identification without forgetting the reciprocal operation known as a re-identification in case when the owner desires to access the information. The efficiency of such approach resides in re-identify data in real time when the owner decides to grow his own data.

Data mining is a powerful discipline that consists of exploring data in order to obtain the unknown knowledge from the known one. This domain presents a powerful tool for data, scientists who are working on data analytics. At the same time, data mining techniques can be utilized by someone to affect other people, for good or evil. This phenomenon has increased, especially with the genesis of big data. For that, a new domain under the name of privacy preserving data mining (PPDM) has seen the light recently. PPDM can be realized as the advanced privacy preserving techniques. This tool aims to protect not only sensitive data that is shown on data sets, but prevents the leakage of sensitive knowledge that can be deducted from analytic processes of data mining. All the same, the major challenge that hinders the advancement of researches resides in the fact that such technique for privacy preserving is efficient if it could obscure the maximum of sensitive data without compromising the usefulness of the information.

This paper covers the problem of privacy preserving over big data. We suggest in this report a new approach for preserving privacy of users by combining access control and data de-identification methods using bio-inspired methods and some mathematical concepts. We first introduce some general concepts used for this approach. Afterwards, we present our approach and its operations. Finally, we present the outcomes and talk about of our experiments by comparing them with some conventional techniques and culminate our work with a conclusion.

RELATED WORKS

Today, privacy preserving is a highly active domain of research. Many works have been done, others are still in progress in order to ensure an efficient protection of users' privacy, especially in the fact of losing the control of data by its owners once it accesses to big data services.

Some of the works are interested in controlling the access to the information. Some of these industrial plants employ a promising technique called Attribute Based Encryption (ABE) such as in (Li, 2013) (Tu, 2012) (Yang, 2011) and (Yu, 2010). In (Crampton, 2009), the author presented a controlling hierarchical access approach using multiple key assignments in cryptography and in which he proposed four iterative schemes: bounded, unbounded, synchronous, and asynchronous in order to make the general idea below the temporal access control. In (Astorga, 2012) the authors demonstrate their new approach of controlling access on a resource-deprived environment in sensor information by integrating the Ladon Security Protocol that offers a secure access using end-to-end authentication, authorisation and key establishment mechanisms in the PrivaKERB user privacy framework of KERBEROS environment. In (Wang, 2006), the authors proposed an approach using Elliptic Curve Cryptography (ECC) to check-up the access to data over sensor networks. They presented their implementation of ECC in TelosB sensor network platform and measured their results by comparing it with the results of (Liu, 2008) and (Malan, 2004). In (Thilakanathan, 2013), the paper has been established to introduce the idea of SafeShare that consists of controlling the access by encapsulation of shared information. Their point of view consists of using the ABE to encrypt, encapsulate, audit and log the data in order to define a perform access control policy. Other works go to the fact of using information content to check the access such as it is pointed out in (Zeng, 2013).

Other works were interested in the de - identification field in which many works have been done for textual data as much as image information.

Concerning textual data de-identification, we can find several works such as the paper presented in (Kerschbaum, 2011) where the authors present an idea that consists of the detection of anomalous events using immune system and graph theory science in order to secure the privacy of users according to their compartments. Other works, that interpret the bulk of works, were interested by medical documents because of its sensitivity. The application of de-identification algorithms in this field was on structured data using terms that are recognized in the medical field. In (Uzuner, 2007) the authors present a state of the art on de-identification of medical records by evaluating some known systems in this topic such as HIPAA as part of the i2b2 (Informatics for Integrating Biology to the Bedside) projects. In (Yun, 2015), the authors propose a fast perturbation technique based on tree structure for database perturbation. Their approach consists of detecting sensitive items within database then represent it in form of tree and perturb it. In (El Emam, 2009), the authors propose a new de-identification algorithm named OLA (Optimal Lattice Anonymization) and evaluate it by comparing its efficiency on medical data with three other algorithms: datafly, Samarati, and Incognito. In (Fernandes, 2013), the authors propose and discuss their new de-identification procedure on mental health records by evaluating the obtained results using this routine on two different datasets: CRIS and MIST.

The image de-identification is no less important than the textual data. As good as the former one, images are widely treated like the work reported in (Al-Najjar, 2011) where the authors suggest a new algorithm for image encryption using logistic map chaotic functions. Their work is compounded on two approaches: changing pixel values without shuffling the image using Pixel Mapping Table and modifying pixel values using rows and column replacement. In (Tong, 2009), as well as in (Al-Najjar, 2011), the authors offer their model of feedback image encryption using chaotic functions and rows and columns controlling based on randomly chosen compound one-dimensional case. In (Upmanyu, 2009), the authors present their resolution of privacy preserving of video surveillance by presenting their point of view consisting of that the person who does the surveillance knows only the terminal output and nothing else. Their paper consists of a demonstration of a framework of a secret sharing scheme based on the Chinese Remainder Theorem. In (Wu, 2014), the writers prove their own model of image ciphering by presenting an algorithm inspired from natural ripple-like phenomenon in order to code images using wave perturbations. In (Yavuz, 2008), the authors look into the efficiency of Time-Reversal (TR) techniques on images ciphering by studying the performance of two algorithms: decomposition of TR-operators (DORT) and multiple signal classification (MUSIC) under the clutter and additive noise schemes. In (Shinzawa, 2014), the authors discuss and analyse the spectroscopic imaging data in multiple-perturbation 2D by proving that this technique can sort out convoluted information about this sort of data. In (Meyer, 2014), the authors offer their new model of denoising of images using the eigenvectors of the Laplacian of the patch graph. They analysed their work using recent results of perturbation of kernel matrices and patch denoising algorithms.

In (Liu, 2011), the authors study the problem of minimal disclosure of privacy with assurance of good functioning of web services with role mechanisms. They modelled the privacy behaviours of services by extending the interface automata to support privacy semantics. In (Lee, 2015), the writers propose an approach of privacy preserving based on feature selection method in order to handle the problem of personalization-privacy paradox in mobile wellness healthcare services.

BASIC CONCEPTS

This section is reserved for the introduction of some basic concepts that we are going to use in our approach.

Mathematical Background

Standard Deviation

This concept gives the measure of dispersion of a specific population starting from its mean which can be regarded as the average of the population's values. It is linearly related to the multiplication of the individuals over the population space; the more the individuals are spread, the higher is the deflection. The following pattern is then applied to estimate the standard deviation of such population of size n .

$$S = \sqrt{\frac{\sum_{i=0}^n (X_i - \bar{X})^2}{n - 1}} \quad (1)$$

Where the $\bar{X}$ denotes the mean that is calculated using the following formula

$$\bar{X} = \frac{\sum_{i=0}^n X_i}{n} \quad (2)$$

And X_i is an individual of the population.

Confidence Interval

It is an inferential statistical measurement that represents an interval of probability in which a population of individuals can fall in according to three essential parameters: the population's mean, the standard deviation and a specific percentage called confidence level. The confidence interval is computed as follows:

$$CI = \text{mean} \pm \text{margin_of_error} \quad (3)$$

Where the margin of error presents the difference between the mean and the extremities of the interval. Two different instances are associated with the equation used to calculate the margin of error: a sample which has a size less than 30 individuals and another which accepts a size more than 30 individuals. The main difference resides in special value called t-value in the first case and z-value in the second one. These two values are drawn from two different Tables as shown in the Table 1 below:

However, the extraction of values from these two Tables is different. In our work, we use z-Table because of the high number of users in big data services. The computation of the margin of error is performed in two steps:

- Extracting z-value from the Table, we compute the $\alpha/2$ value where α is the confidence level. Let's get an example of confidence level of 90%. The $\alpha/2$ value is $0.90/2 = 0.45$. After that, we look for the closest value in the Table, we find 0.4495 and 0.4505. Then for each one of these values we calculate the corresponding row and the corresponding column and we get 1.64 and 1.65. Finally the z-value equals to $(1.64 + 1.65) / 2 = 1.645$
- Now we have the z-value, the margin of error is calculated using the following formula:

$$\text{Error_margin} = \text{z-value} \times \frac{S}{\sqrt{n}} \quad (4)$$

where the S is the standard deviation and the n is the size of the sample

Table 1. A samples from t-Table and z-Table

Z	0.00	0.01	0.02	0.03	0.04	t	0.40	0.25	0.10	0.05	0.025
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	1	0.324920	1.000000	3.077684	6.313752	12.70620
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	2	0.288675	0.816497	1.885618	2.919986	4.30265
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	3	0.276671	0.764892	1.637744	2.353363	3.18245
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	4	0.270722	0.740697	1.533206	2.131847	2.77645
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	5	0.267181	0.726687	1.475884	2.015048	2.57058
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	6	0.264835	0.717558	1.439756	1.943180	2.44691
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	7	0.263167	0.711142	1.414924	1.894579	2.36462
0.7	0.2580	0.2611	0.2642	0.2673	0.2704	8	0.261921	0.706387	1.396815	1.859548	2.30600
0.8	0.2881	0.2910	0.2939	0.2967	0.2995	9	0.260955	0.702722	1.383029	1.833113	2.26216
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	10	0.260185	0.699812	1.372184	1.812461	2.22814
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	11	0.259556	0.697445	1.363430	1.795885	2.20099
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	12	0.259033	0.695483	1.356217	1.782288	2.17881
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	13	0.258591	0.693829	1.350171	1.770933	2.16037
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	14	0.258213	0.692417	1.345030	1.761310	2.14479

a) A sample from Z-Table

b) A sample from T-Table

Primitive Root

In informatics security the primitive root is an important concept used in several cases, particularly in the case of sharing the keys in public key cryptography systems. Formally a primitive root r of a number P is the number that satisfies the following attribute:

r is a primitive root of $P \Rightarrow i, j \in \mathbb{N}$, if $i \neq j$ then $r^i \bmod P \neq r^j \bmod P$

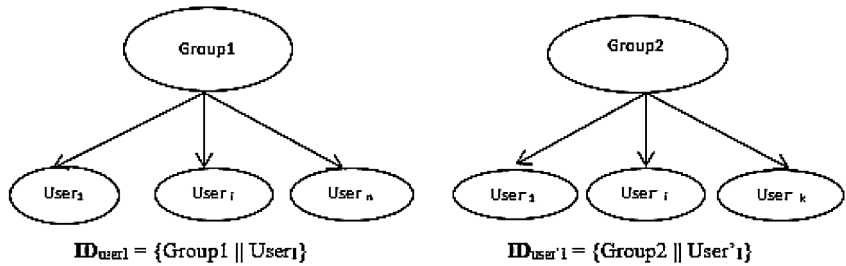
Nevertheless, in mathematics there is no accurate way to compute a primitive root of a number. Instead, there is a method to verify if such number r is a primitive root of a number P if for each prime factor f that compound the Euler's indicator of P $\ell(P)$: $r^{\ell(P)/f} \bmod P \neq 1$ $r^{\ell(P)/f} \bmod P \neq -1$ in the same time. The most known algorithm which uses the primitive root is the famous Diffie-Helman algorithm for sharing secret keys and known as discrete logarithm problem. It was proved in this algorithm that for a number r being primitive root of a number P , if we know r , P , and the result of $r^a \bmod P$ we can never deduct the number a .

Hierarchical Identification

The big data concept comes out with a real development not merely in terms of data volume, but likewise in term of number of users which makes the identification of them, a crucial problem leading to the search for new technics of identity selection. One of these technics is a promising and a new method called Hierarchical Identification that aims to profit from different information associated to the users and on which the identities are based using the chain process as shown in the Figure 1:

This method delivers a major advantage which resides in the ability of using the same identifier for multiple users in different groups, allowing the identification of a large number of users with small size of identities. It can be useful in many cases that are linked to the identification process such as the authentication mechanisms.

Figure 1. Hierarchical identification



Taxonomy of Data Perturbation Techniques

Data modification is at present a widely known domain of research in the security domain, especially with the grown amount of shared data over the network. This section presents a general view of the perturbation techniques. Data modification techniques can be grouped in two major classes as figure 2 shows below:

Value Based Perturbation

Noise Additive Perturbation Techniques

Noise addition techniques are the most used ones in research area. This kind of techniques represents a lot of techniques that consist of adding or multiplying several stochastic or randomized numbers to a quantitative confidential data in order to protect the confidentiality of these last. The figure 3 indicates the general view of data perturbation using noise addition, according to (Mivule, 2013).

Randomized Response Technique

Is a technique that consists of dealing with a structured survey interview in which the purpose is to respond a sensitive issue while maintaining the confidentiality? This technique was introduced

Figure 2. Data modification techniques

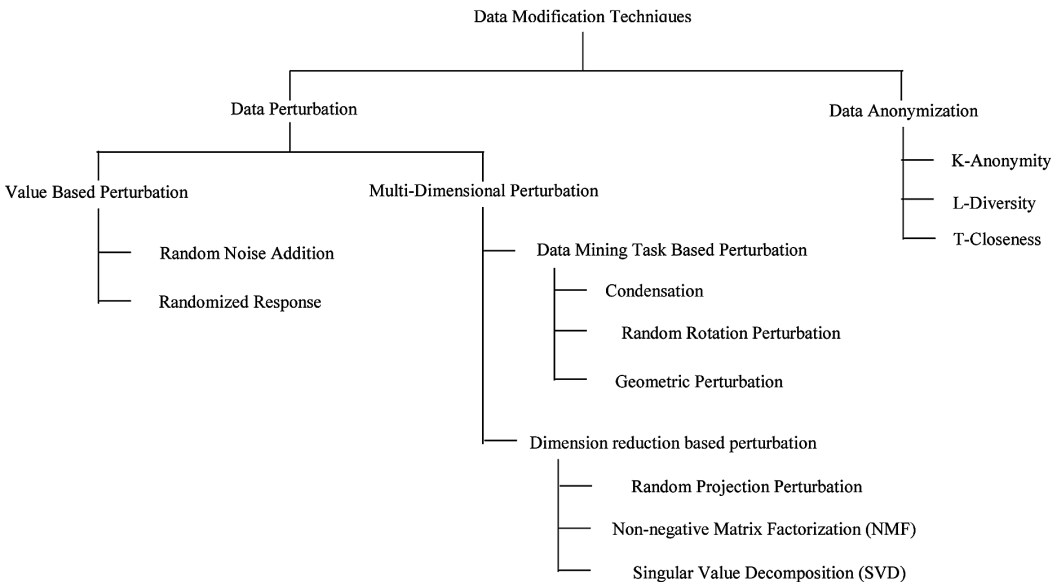
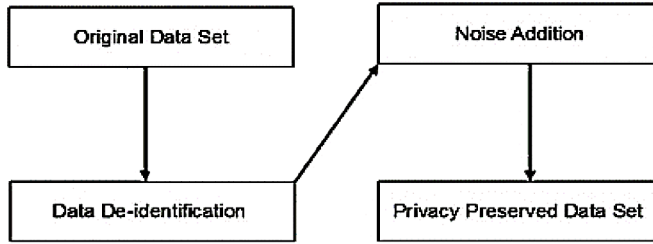


Figure 3. General framework of data perturbation using noise addition (Mivule, 2013)



by S. L Warner in 1965 and modified later by Greenberg in 1969. Formally, for a respondent who receives an attribute A, let P be the probability of A being sensitive, so that the other side will have a probability of 1-P. In order to avoid any revealing of information about the selected statement to the interviewer, an elementary probability is used to get an estimation of proportion named π of A in the population as follows:

$$\lambda = P \lambda + (1-P) (1- \lambda) \quad (6)$$

Multidimensional Perturbation

Data Mining Task Based Multidimensional Perturbation

Condensation based Perturbation: This technique consists of protecting the privacy by preserving the covariance matrix of multiple columns. It works by partitioning the dataset into k groups. After that, for each group a random record is selected, then regarded as the core of the group. The technique next chooses (k-1) nearest records to be the other members. Finally, the system removes the chosen k records from the data set before forming some other group. This technique aims to preserve the eigenvectors and eigenvalues of the dataset

Random Rotation Perturbation: This technique consists to choose randomly a set of records and so applying the perturbation by rotating or transforming randomly the data. To answer that, the data are mapped by a matrix $X_{n \times m}$ then a random orthonormal rotation matrix R is generated. Finally, the perturbed data equals to the product of R with X. Many approaches are published in this kind of techniques using classification and transformation techniques such as SVM, KNN, K-means, Inner product and kernel methods.

Geometric Perturbation: The geometric based models are a combination between rotation, transformation and noise addition techniques. Formally, let's consider $X_{n \times m}$ a dataset matrix with n columns and m records, $R_{d \times d}$ a random rotation orthonormal matrix, T a translation random matrix and $N_{n \times m}$ a random noise matrix. The geometric perturbation, then is the results of the following formula (Patel, 2013) and (Chen, 2011):

$$G(X) = RX + T + N \quad (7)$$

In summation, the geometric perturbation known by her invariance against geometric modification. However, this kind of techniques guarantees more privacy rather than random rotation and condensation based techniques.

Dimension Reduction Based Model

Random Projection Perturbation: Introduced by Johnson-Lindenstrauss, this technique is based on the use of a random matrix to project the original dataset into a low-dimensional subspace. Formally, consider $\epsilon > 0$ and $0 < \delta < 1/2$, let k be a positive integer in which $k \geq k_0 = O(\epsilon^{-2} \log(1/\delta))$. For each dataset P in $\mathbb{R}^d$ there is a function $\varphi: \mathbb{R}^d \rightarrow \mathbb{R}^k$ such that for all $x, y \in P$:

$$(1 - \epsilon) \|x - y\|_2^2 \leq \|\varphi(x) - \varphi(y)\|_2^2 \leq (1 + \epsilon) \|x - y\|_2^2 \quad (8)$$

So that this lemma is conducted on the fact that n points could be projected from $\mathbb{R}^d$ to $\mathbb{R}^k$ while preserving the Euclidean distance between points within an arbitrarily small factor.

Nonnegative Matrix Factorization (NMF): The NMF technique refers to the use of non-negative matrices and is formulated as follows: given a matrix V of size $n \times r$, NMF consists of finding an approximate factorization $V \approx WH$ where W and H are non-negative matrices of size $n \times m$ and $m \times r$ respectively. The product WH can be seen as the compressed form of V . Even so, the optimality of the algorithm resides in the choosing of W and H matrices in such a manner that boils down the fault rate between V and WH . Several error functions have been proposed in this area, but the most known and used ones are two major functions that are the squared error (Euclidean distance) presented by the function (9) and K-L Divergence presented by the function (10).

$$E(W, H) = \sum_{ij} \left(V_{ij} - (WH)_{ij} \right)^2 \quad (9)$$

$$D(A||B) = \sum_{ij} \left(A_{ij} \log \frac{A_{ij}}{B_{ij}} - A_{ij} + B_{ij} \right) \quad (10)$$

Where A is the matrix V and B is considered as WH . Thus the NMF requires that all the components of W and H must be non-negative.

Singular Value Decomposition (SVD): This technique is one of the most known techniques of privacy-preserving data mining. It provides a perturbed data so that the user will be accused of extracting original data patterns. Formally, given a matrix A of size $n \times m$ where n is the number of information objects and m is the number of properties. The SVD of a matrix A is set by the following function:

$$A = USV^T \quad (11)$$

Where U is an orthonormal matrix of size $n \times n$, S is an $n \times m$ diagonal matrix in which the diagonal entries are nonzero and equal to the Rank (A) in a descending order. Rank (A) is the rank of the matrix A , and V^T is an $m \times m$ orthonormal matrix. However, the perturbed data of this technique is computed as follows:

$$A_k = U_k S_k V_k^T \quad (12)$$

Where U_k is an $n \times k$ matrix that contains the first k columns of U , S_k is a $k \times k$ diagonal matrix that holds the k largest nonzero diagonal entries of S , and V_k^T is a $k \times m$ matrix that contains the k first rows of V^T . The privacy of this technique resides in the number k . The more k is increased, the more amount of original data is kept. The increasing rate of k improves the utility of data at a cost to privacy.

Data Anonymization

K-Anonymity

This technique was introduced by Latanya Sweeney in 2002. The purpose of here paper was to resolve the problem “Given person-specific field-structured information, produce a release of the data with scientific guarantees that the people who are the subjects of the data cannot be re-identified while the data remain practically useful”. Formally, let RT be a table with n attributes $(A_1 \dots A_n)$ And QIRT the quasi-identifier (Sweeney, 2002) corresponding to RT , such system is called k -anonymity if and only if for each sequence of values in RT [QIRT], it appears at least k time in RT [QIRT]. There are two major categories of methods to achieve a k -anonymity process:

Suppression: The attributes’ values are replaced by an asterisk ‘*’. In this method some or all the values of a sensitive attributes are replaced.

Generalization: In this method the values are replaced by a broader category. For example, the age attribute can be replaced by intervals. 19 replaced by <20 , 25 replaced by $20 < 30 \dots$ etc.

L-Diversity

This technique represents an extension of k -anonymity technique. In fact, the l -diversity handles with some weaknesses of k -anonymity. The purpose of this technique is to reduce the granularity of data representation even if that came at a cost of effectiveness and data utility. Formally, to have an l -diversity model is that if there are at least l “well presented” data of a sensitive attribute. For a given table that holds q -block sets of sensitive values, we say that the table is l -diverse if every block in it is an l -diverse. Nevertheless, (Machanavajjhala, 2007) defines the expression “well presented” in three ways:

Distinct l-Diversity: At least there are l distinct values of the sensitive attributes in each equivalence class

Entropy l-Diversity: Given E the entropy of equivalent class and equals to the negation of summation of s across the orbit of the sensitive attribute of $p(E, s) \log(p(E, s))$ where $p(E, s)$ is the fraction of records in E that have the raw values. For such table T , we said it has entropy l -diversity if for each equivalence class c

$$E(c) \geq \log(l) \quad (13)$$

Recursive (c-l)-Diversity: The most common value does not appear too often while the less common ones appear more frequently

T-Closeness

T -closeness is an amelioration of l -diversity against its known attacks. This technique treats the distinct values of sensitive attributes by bringing into account the distribution of these values. An equivalence class is supposed to be t -closeness if and only if the distance between the distribution of sensitive attribute in this class and the dispersion of the attribute in the whole table is never more than a threshold t .

OUR APPROACH

We bring in suggesting a new approach consists of combining the access control and data perturbation techniques. This part is directed to the general presentation of our organization. This system is compound on two major steps: defining access rights in which the digital signatures and the access control matrix are computed, and the control of access to data in which the perturbation techniques take place in order to protect sensitive information from illegal access. Our system uses four new perturbation techniques: the first is based on an evolutionary method known as the immune system while the second one is based on stochastic methods known as iterated local search which is made in choice with the third one which is a bio-inspired firework algorithm based algorithm. And the fourth technique is a new bio-inspired algorithm from the natural phenomenon of apoptosis.

Access Control

The Figure 4 shows the main process of the first step in our system

As figure 4 indicates above, the first process of our system consists of computing access rights of the users of the data. This process works as follows:

1. First of all, an identification step consists of computing the identity of each user in his group. For that we use a hierarchical identification in which the identity of each user in his group will be the result of the concatenation of the group's identity and the real user's identity.
2. Since the approach is controlled using users' identities, it is significant to hide these identities during the building of an access control matrix. To answer that we use a previous homomorphic scheme that we published in (Rahmani, 2014) in which we utilize the functionalities of crossover and mutation of genetic algorithms to enhance a cryptographic scheme called TSZ (Chabanne, 2005). The aim of this step is to benefit from the homomorphic advantage in order to hide real identities and preserving its utility in the system. In this system we use the same cryptographic key for all the groups in which our system computes the key, then share it with the users who are responsible of ciphering their own signatures.
3. The third step consists to normalise the ciphered identities in order to use it to compute confidence intervals for each group.
4. This step can be run in parallel with the step 3. In this step, our system computes the digital signatures for users using the primitive root. For each group, the system generates a big prime P and its primitive root r . After that, the digital signatures of the users that belong to the group are the upshots of the equation 14 bellow

$$\text{Sign}(u) = r^{id_u} \bmod P \quad (14)$$

5. The final step consists to sign data by its owners' signatures. After that, the system computes the access rights as follows: if the user identity does not belong to the confidence interval of the group then he will have no access right to the data. Else if the signature corresponding to the user equals to the one used to sign the data he will have full access to it else he will have only read right.

The result of this step is a matrix (user x data) in which each line defines the access right of a user to all the data and each column defines who has access and which right has each user of this data.

The second step of our system consists to verify the access right to a data once a user tries to get in. The figure 5 bellow indicates the process of this step.

As we notice in the figure 5, once the access control matrix is computed it is stored on a secured place in the system. When a user wants to access to data a verification procedure will verify the access right corresponding to the user and the data (line and column) so that if the user has access to it the

Figure 4. Process of definition of access rights

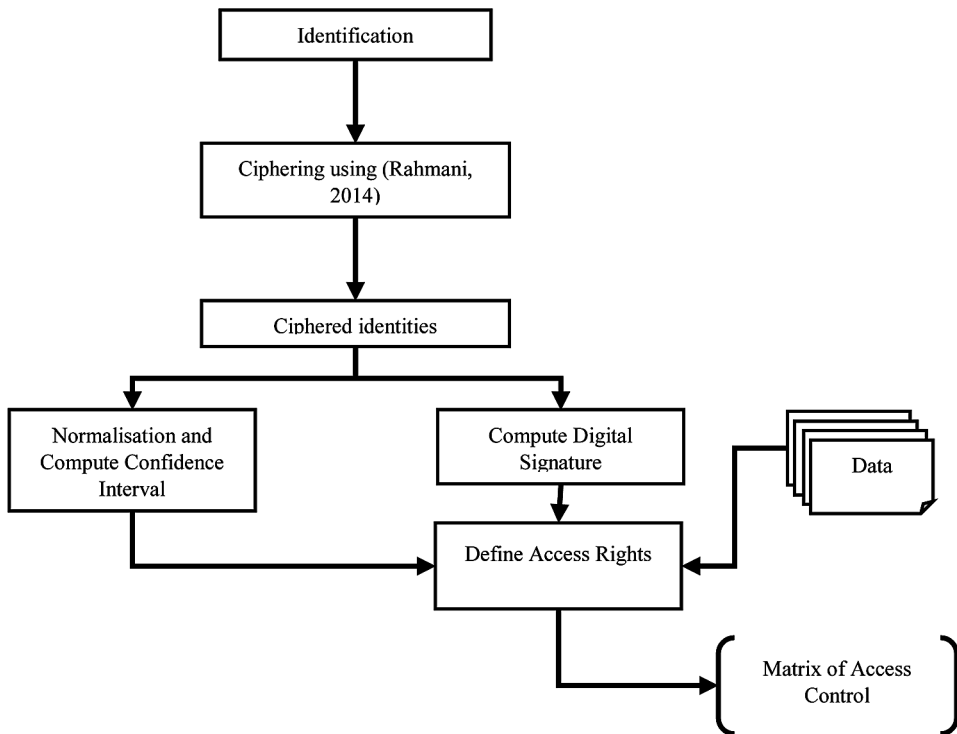
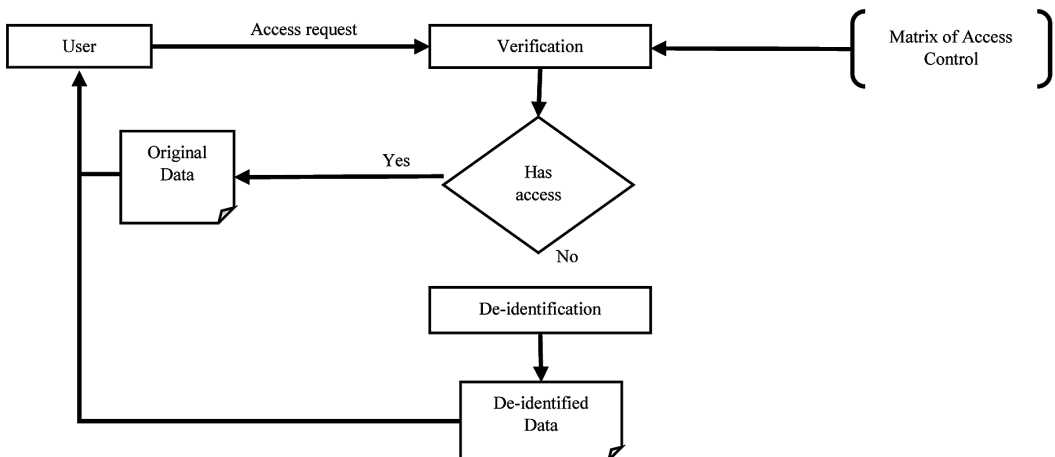


Figure 5. Controlling the access to the data



system send the original data as it is to him. Else the data will pass first by a de-identification process in which the sensitive information is detected and replaced by others. The result of a de - identification process is a temporary de-identified data which will be transmitted to the user.

De-Identification

Concerning the de-identification process, we have utilized different bio-inspired methods for each type of data:

- The immune system which is used for detection and removing of sensitive information within unstructured textual information
- We have used iterated local search (ILS) in choice with firework algorithm for image data.
- And in the last, we have proposed a new bio-inspired technique from the natural phenomenon of cell changing in human bodies known as apoptosis.

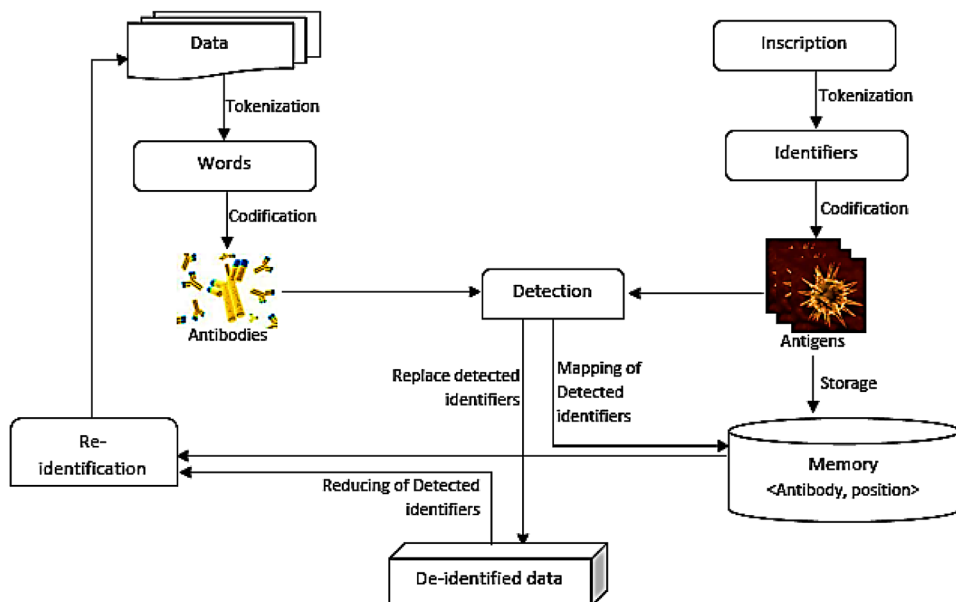
In the remainder of this section we will handle the de-identification of the data by type.

De-Identification of Unstructured Textual Data

This step uses the personal information of the user using the approach presented in (Rahmani, 2015) (1). The figure 6 bellow presents the general idea of perturbing textual data:

As showing in figure 6, the way to do that is deceptively simple. Each word of the personal data is changed into a vector of ASCII codes corresponding to the letters that compose the word. Then this vector is taken as an antigen. After that when a text comes to be de-identified, it will be tokenized into words, then the quarrel will be transferred into vectors of the ASCII codes as well as the personal information. The words, compound the data are considered as antibodies. Subsequently, for each antigen, our system computes the distance between it and all the antibodies. Using a pre-initialized threshold, the word is considered an identifier if his distance with the antigen is less than the threshold. At one time an identifier is found, it will be stashed away in a memory space reserved for the user. The way that identifiers are stored in the memory is similar to the map process of mapReducing where the couple <key, value> is the couple <identifier, code of the word> where the code of the word is a special code formed from the number of the textual document concatenated with the position of the word in this file. After detecting and storing the identifiers, an iterative process starts in order

Figure 6. De-identification of unstructured textual data



to replace these identifiable words, the system choose randomly two identifiers from the memory. Then it concatenates them and for a set of iterations, two random letters from the result word will be chosen and permuted. At the end, a random position will be chosen in order to split the word into two parts. The first part will replace the first identifier and the second part will replace the other one. For future texts, the detected identifiers of past texts are considered also as antigens.

However, our system takes in account ones the owner wants to get his text files. In this case a re-identification process takes place using the memory of the user. This process is similar to reduce process of mapReduce principle and consists of reading the code of the word, then if the number of the document is the same as the treated one, the system replaces the word in the indicated position in the code with the key word of the couple.

De-Identification of Image Data

In our system, the image data are treated differently from the textual one. To de-identify image data, we use two approaches that we proposed in (Rahmani, 2015) (2) using bio-inspired firework algorithm and a stochastic iterated local search algorithm (ILS).

Referable to the perturbation and permutation functionalities of ILS, this process consists to divide the images into components (picture elements, lines or columns) in which each part stands for a vector. Nevertheless, we present the pixels of the images by ARGB presentation. Once the individuals are created, the ILS algorithm is used in order to perturb the image using an iterative random permutation process.

The Figure 7 illustrates the mechanism we set up to explore ILS algorithm in our organization.

By observing the figure 7, we can describe the process of perturbing image data with ILS algorithm as follows: first of all, the system consists of dividing the picture into parts (pixels, lines, or columns). After that, our system executes a random permutation between the parts in purpose to get a disorder of the picture. And since, as we mentioned before, the picture is presented by the vector of ARGB of its pixels, a local search is used using distance measurement in order to get the set of parts that are further from each other. After getting the sets, the system consists of executing iteratively a set of permutation and local search process until it reaches maximum number of iterations. At the end, the system provides a perturbed image which is only the original image with pixels that their positions are changed.

Artificial Firework Algorithm (FWA) is another approach that we use in this system. This algorithm presents a good model for perturbation of image data regarding to the natural process of explosion of fireworks. Figure 8 shows the general idea of using FWA in image perturbation:

The figure 8 above shows the perturbation process of image data using FWA. This process can be resumed in the following points:

- First of all, the image is codified pixel by pixel using their positions
- Next, a random number n of pixels is selected for next steps (n is also a random number that variate in each iteration between 1 and half number of pixels)
- Each one of the selected pixels will hold a firework that will explode and generate sparks
- The next step consists of computing the number of sparks of each exploded firework
- Then for each spark, the system computes his amplitude using distance computation then get the pixel at the position in which the amplitude id added to the position of the firework
- Next, the system shifts between the pixels at the firework position and the one at the end of the amplitude of the spark

This process continues working in iterative way until a stop criterion is reached. The stop criterion can be either a predefined number of iterations or simply a comparison of the correlation of adjacent pixels of the image at the entry of the iteration and the one at the end of the iteration. If we use the correlation comparison criterion we must ensure that the correlation of finale perturbed picture must be as close as much to the 0.

Figure 7. Image perturbation process using ILS algorithm

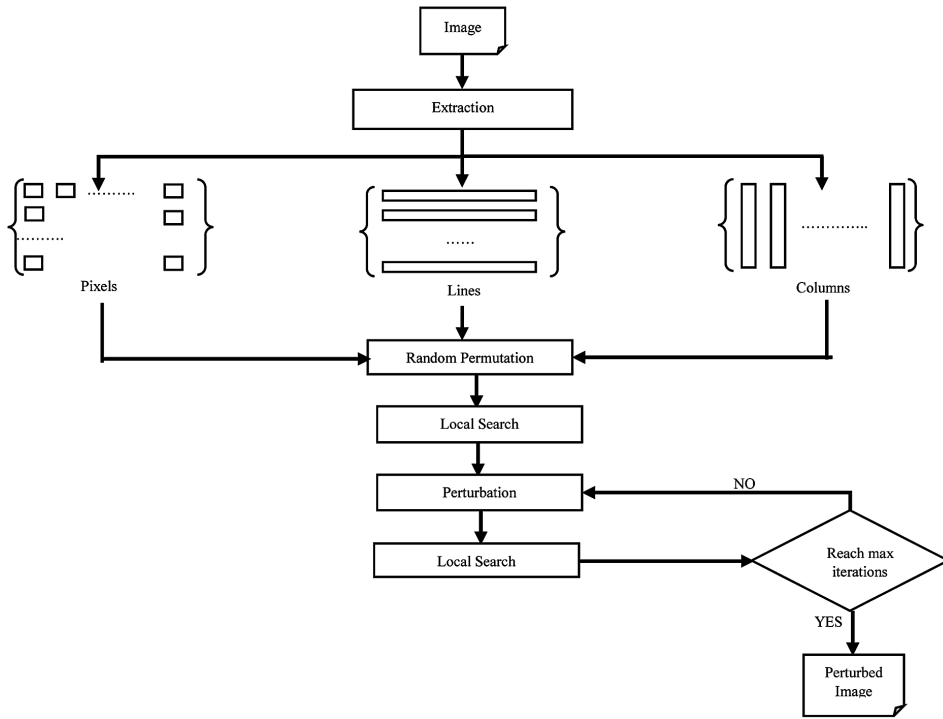
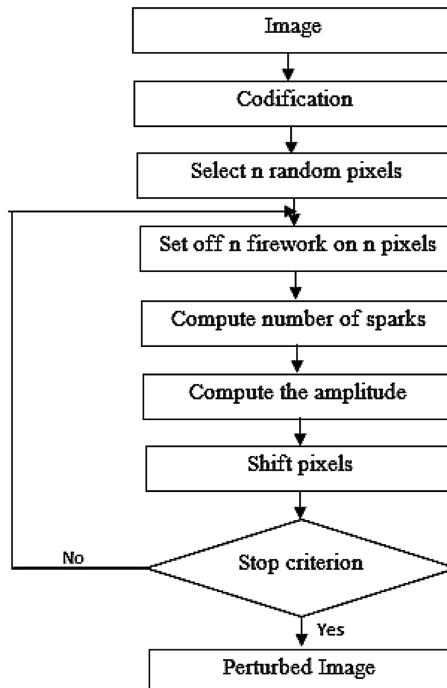


Figure 8. Perturbation of image data using FWA



De-Identification of Structured Data

In this part we have proposed another approach using a new bio-inspired algorithm from the natural phenomenon of apoptosis known by cells' suicide in human being anatomy. This approach is presented in (Rahmani, 2015) (3). Figure 9 shows the process of hiding sensitive information in structured data.

By regarding the figure 9, we resume the process of hiding sensitive information within structured data as follows: at the beginning, the system consists of isolating the sensitive attributes such as names and social numbers. The isolation process can be either done manually while creating the data base or using algorithms for that such as data mining and feature selection methods. The isolation step is inspired from the isolation of diseased cells. After that, as well as in natural apoptosis process where the core of the isolated cell starts dividing into small pieces, our system starts the budding process that aims to divide the isolated attributes into two groups according to their types (numerical or nominal). The budding step is much simple and consists only on reading the type of the attribute without any complex method. Next, we have defined two algorithms of perturbation each one for a group of attributes.

For numerical attributes, we have defined a perturbation method using noise addition based on Box-Muller formula for Gaussian Random Number Generation (Thomas, 2007). The Box-Muller based perturbation method consists of reading a pair of sensitive values (x, y) in order to perturb the x value (the y value is used as intermediate value and chosen randomly from the rest of the values in the same attribute) and generating another pair of numbers (a, b) where $a = \sqrt{-2 * \log y} \bmod x$ and $b = 2 * \pi * y \bmod x$. after that, our system computes a noise using a and b where another pair of numbers is generated (r, r') where $r = a \sin(b)$ and $r' = a \cos(b)$. At the end if the r' is less than 0.5 the noise r is added to the x else the noise r is subtracted from the x.

Concerning nominal attributes, we have defined a different way of perturbation. To perturb a certain word, our system transfers it first into a vector of ascii codes, let's consider it A. Then, it choses randomly another word and transfer it into a vector, let's call it B. After that if the length of A and B are different, the system completes the smaller one by zeros (Blanc character). Next, our system uses the equation 14 bellow to generate the perturbed vector:

$$D(A||B) = \left(A_{ij} \log \frac{A_{ij}}{B_{ij}} - A_{ij} + B_{ij} \right) \quad (14)$$

At the end, the new vector D(A||B) is transferred into a word that will replace the word A.

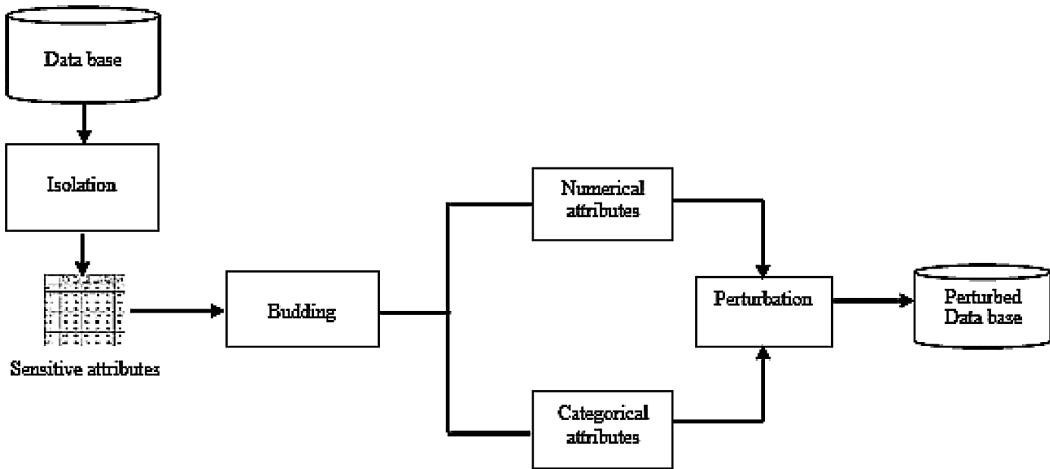
EXPERIMENTS AND RESULTS

We carried on a set of experiments by building a framework compound on 10 groups; each one contains 2000 user. For the experiments we use some collected data sets to contain different data of different sizes and different formats. Our data set is compound on 20000 unstructured textual file, 1000 records on database, and 5062 images.

For the evaluation of our results we based on several criteria for each part of our system in which we varied some parameters. We use time of generation and verification of signatures, and access rate to evaluate access control part. Correlation coefficient and de-identification time to evaluate the de-identification of image data. And time of de-identification and success rate for evaluation of de-identification of textual data.

This section is addressed to the representation and discussion of different results that we got from the experiences.

Figure 9. De-identification of structured data



Access Control

As a beginning, we evaluate the access control model by evaluating the time of generation and verification of signatures as well as the table 2 and figure 10 shows bellow. The results of this part are compared with other works presented by TinyECC system of (Liu, 2008) and ECC-AC of (Wang, 2006).

As shown in the results previously presented, it is clearly noticed that our system took less time than the other two systems with 0.107 seconds for generating one signature including the identification, encryption of identity and computation of digital signature. While ECC-AC presents the worst results by taking 8.24 seconds. However, the results indicating the time of verification of a signature once a user decides to enter into a data is not different from the generation one.

Another measure we apply to evaluate access control model is the rate of success of accessing to data resulted from each group in term of choosing confidence level. After trying several confidence levels, we choose the best results to be presented in this part. Yet, before presenting the results, the table 3 shows the corresponding z-value of choosing confidence levels.

The Table 4 indicates the results of access rate presenting by percentage.

The results presented in table 4 and figure 11 show a high efficiency in concerning the success of accessing for authorised users to data. However, the best results are presented by the groups 5, 6, 7, and 8 in which we observed a full efficiency with 100% of authorised users could access to their data and 100% of unauthorised users couldn't access to it. Meanwhile, the other groups indicate a disparity of results depending on the chosen confidence level and that due to the distribution of users' identities over the groups. For the groups that exemplify a high efficiency, the identities were too near to each other and closer to the sum of the margin of the group. That's why we needed to the less value

Table. 2. Comparison of time of generation and verification of signatures

Approach	Time of generation of signature (seconds)	Time of verification of signature (seconds)
Our approach	0.107	0.149
ECC-AC	3.35	6.78
TinyECC	8.24	16.28

Figure 10. Results of comparison of time of generation and verification of signatures

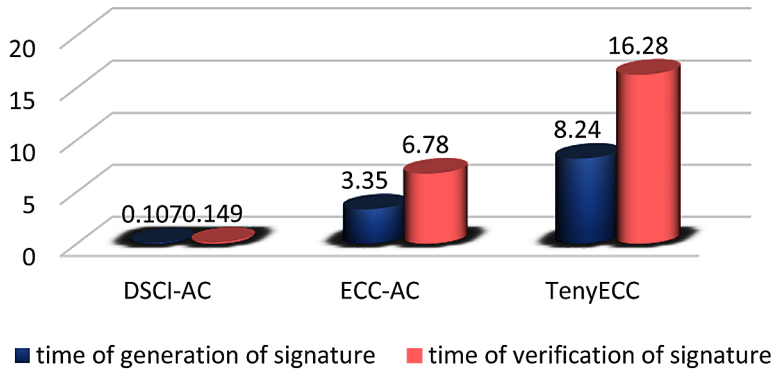


Table. 3. Corresponding z-value for each chosen confidence level

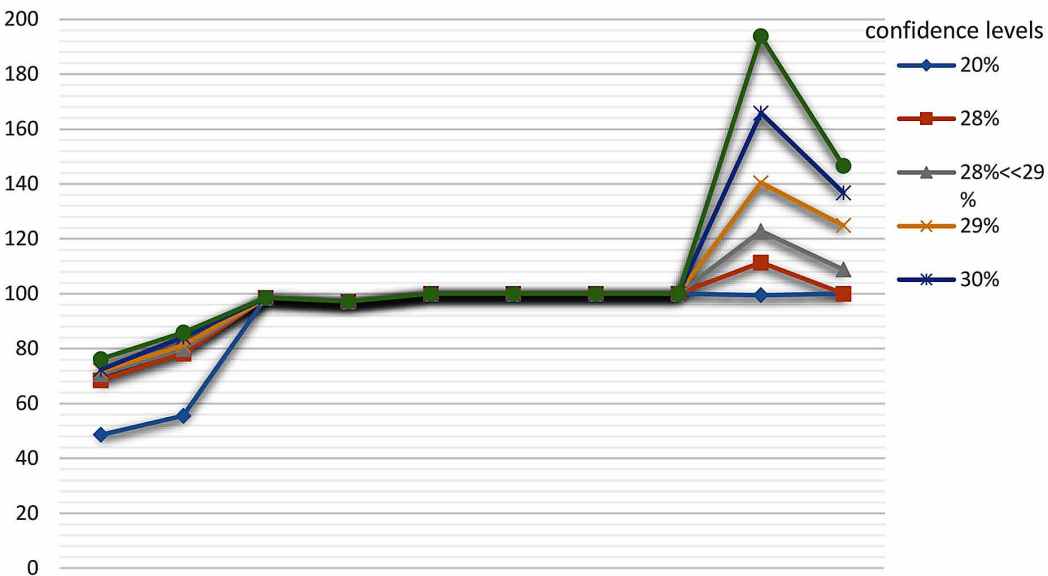
Confidence interval	20%	28%	28% < 29%	30%	31%
z-value	0.255	0.355	0.365	0.375	0.385

Table. 4. Access rate by group in term of chosen confidence level

Confidence level Domain	20%	28%	28% < 29%	29%	30%	31%
Group 1	48.62	68.51	70.72	71.82	72.36	76.24
Group 2	55.55	78.20	79.91	81.20	84.19	85.90
Group 3	98.54	98.54	98.54	98.54	98.54	98.54
Group 4	97.13	97.13	97.13	97.13	97.13	97.13
Group 5	100.00	100.00	100.00	100.00	100.00	100.00
Group 6	100.00	100.00	100.00	100.00	100.00	100.00
Group 7	100.00	100.00	100.00	100.00	100.00	100.00
Group 8	100.00	100.00	100.00	100.00	100.00	100.00
Group 9	99.48	111.39	122.80	140.41	165.80	193.78
Group 10	100.00	100.00	108.81	124.87	136.79	146.63

of confidence level to demonstrate the efficiency. Nevertheless, in the case of group 10, the identities were too near to each other, but too far from the middle of the margin and that's why the minimum confidence level could lead to the overabundance of the margin, by consequently, allow for a big act of unauthorised users to access to information. The case of groups 1 and 2 is entirely different, the identities in this case were largely distributed in the group margin in which we have needed for the maximum of confidence level in order to cover the maximum of authorised users.

Figure 11. Access rate by group in term of chosen confidence level



De-Identification of Unstructured Textual Data

The next part that we evaluate is the de-identification part. For that we divide our evaluation of three major criteria: time of de-identification of each algorithm, the success rate of de-identification of textual data, and correlation coefficient for de-identification of image data.

For textual data, we apply different parameters by utilizing the system for all textual data including structured and unstructured one:

- The distance measure used for detection, for that we use five different measures: Euclidean, Tchebychev, Manhattan, Cosinus, Minkowski.
- For each distance we use five different thresholds varied from distance to another.

For the evaluation of this part we use different measures: the number of detected identifiers and removed ones counted by words, also the time of de-identification counted by seconds for each distance. The Table 5 show the best results in term of choosing threshold for each distance.

Table 5. Results of de-identification of textual data

Distance Measure	Best threshold	#detected identifiers	#removed identifiers	Time of de-identification
EUCLIDEAN	130	2 530	2 313	280.413
MANHATTAN	100	2 569	2 357	70.003
COSINUS	120	2 574	2 246	76.0034
MINKOWSKI	100	2 672	2 469	209
TCHEBYCHEV	150	2 627	2 499	77.2002

By analysing the results presented in Table 5, we observed that the TCHEBYCHEV distance shows the optimal results among the tested ones with maximum of detecting identifiers (an average of 2 627 detected word) with an acceptable time of de-identification (an average of 77.2002 seconds) for large datasets (20000 unstructured texts and 1000 database records). Meanwhile, the EUCLIDEAN distance shows the worst results with less detected identifiers (2 530 detected word) and biggest time of computing (an average of 280.413 seconds). Concerning the other distances, it presents a satisfactory result. Some of it in term of time of computing, such as MANHATTAN distance which present the best time of computing with an average of 70.003 seconds but less detected identifiers. And others in term of detected identifiers such as MINKOWSKI distance which presents the best results with an average of 2 672 detected terms, but more time of computing with an average of 209 seconds.

For more credibility of evaluation of this part; we compare our best results in term of time of responding with the systems: FPUTT presented in (Yun, 2015), and HHUIF and MSICF presented in (Yeh, 2010).

The results presented in Table 6 revealed that the system FPUTT proposed by Yun gives the best result with about 115 seconds for data perturbation; while our system gives a satisfactory result with almost 143 seconds. Our system presents better solution than the HHUIF and MSICF systems in term of time of execution.

De-Identification of Image Data

The third part that we analysed during our experiments is the de-identification of image data. In this case, and since ILS algorithm fit into the family of multi-stage optimisation algorithms, we use two ILS algorithms. One using two stages (ILS-2opt) and other using three stages (ILS-3opt). However, the evaluation of this part is based on two criteria: time of de-identification and correlation coefficient of original and de-identified image. The Table 7 indicates the time of de-identification of the both algorithms using different images with different sizes and different ways of segmentation like the figure 10 indicates above.

The results shown in Table 7 revealed that our approaches present a satisfactory time of de-identifying of images. However, by analysing these results, we can clearly notice that dividing the images into lines or columns takes less time than the one using pixels, and that is due to the number of generated parts. We noted that the use of ILS-3opt algorithms with lines gives less time than all the other methods with an average of 9.311 seconds, while ILS-2opt using pixels dividing method takes longer than the others with an average of 18.092 seconds. On the other hand, we noticed that the picture's dimensions play an important role in this case because of the number of objects that each one could generate. The smallest are the images the more are easy to be treated and hidden. Furthermore, comparing by pixels dividing of the three methods, we have notices that firework algorithm took less time than the others with 14.839 seconds as average of perturbation of several image sizes comparing to 15.176 seconds for ILS-IP-3opt and 18.092 seconds for ILS-IP-2opt. the reason that there is no results of FWA-IP with columns and lines dividing method is because these two will limit the direction of sparks into two directions (up or down) or (right or left) which contradicts the principle of the algorithm.

Since the most of algorithms consist of perturbing pixel values, we will in the newt of this part of evaluation interest only by pixels dividing methods for ILS-IP perturbation.

The Figure 12 below illustrates a sample of execution of our perturbation model, (Tong, 2009), (Al-Najjar, 2011), (Gu, 2014), and (Chen, 2015) applied to the famous Lena image.

From Figure 12 above, we noticed that the other works, (Tong, 2009), (Gu, 2014), (Chen, 2015), and (Al-Najjar, 2011), consist of changing pixel values which could affect other operations in the future, such as retrieval models. Meanwhile our system consists only of perturbing pictures by shuffling pixels in the image itself, in other words, our system uses pixels values such as ARGB or grey level only for distance computation between pixels in order to avoid the permutation of nearly

Table 6. Comparison of time of responding of our system, FPUTT, HHUIF and MSICF

Model	Time of Response (seconds)
Our system	70.003
FPUTT	114.946
HHUIF	262.469
MSICF	431.127

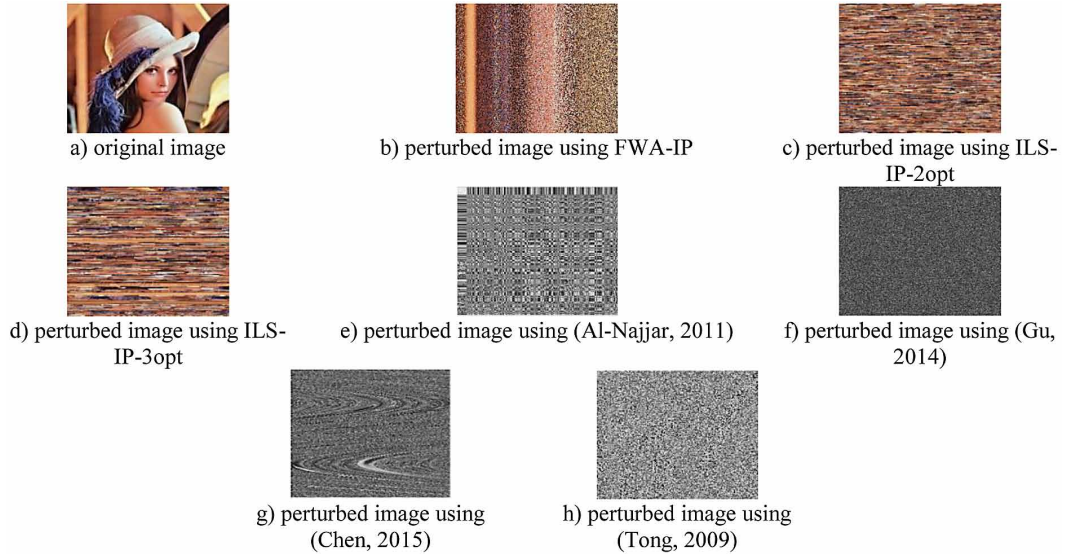
Table 7. Time of de-identification of images using ILS-2opt and ILS-3opt

System		Image Size	Time of answer (Seconds)	Average of time of de-identification
FWA-IP		320 x 240	05.977	14.839
		384 x 288	07.315	
		768 x 576	31.224	
ILS-IP-2opt	Pixels	320 x 240	08.003	18.092
		384 x 288	11.851	
		768 x 576	34.423	
	Columns	320 x 240	04.306	09.907
		384 x 288	05.836	
		768 x 576	19.580	
	Lines	320 x 240	04.390	10.398
		384 x 288	05.957	
		768 x 576	20.848	
	Pixels	320 x 240	06.194	15.176
		384 x 288	08.206	
		768 x 576	31.130	
ILS-IP-3opt	Columns	320 x 240	04.287	09.880
		384 x 288	05.437	
		768 x 576	19.916	
	Lines	320 x 240	04.180	09.311
		384 x 288	06.203	
		768 x 576	17.551	

similar pixels. For the evaluation of this model compared to the other models we have confronted a comparison in term of correlation coefficient defined in (Tong, 2009) by the following formula:

$$\text{Corr} = \frac{N \times \sum_{j=1}^N (x_j \times y_j) - \left(\sum_{j=1}^N x_j \times \sum_{j=1}^N y_j \right)}{\sqrt{(N \times \sum_{j=1}^N x_j^2 - (\sum_{j=1}^N x_j)^2) \times (N \times \sum_{j=1}^N y_j^2 - (\sum_{j=1}^N y_j)^2)}} \quad (15)$$

Figure 12. Result of perturbation of Lena image



Where x_j and y_j are two adjacent pixels and N is the number of chosen pixels. The correlation takes values from -1 to +1. The more it is close to 0 the more the pixels are nonlinear related to each other and the less the image is significant. And the more is close to -1 or +1 the more the pixels are linearly related to each other and the more the image is significant. However, in our experiment we use two correlation coefficient: horizontal one and vertical one. Table 8 indicates the results of this experiment.

By analysing the effects shown in Table 8, we notice that the fact of keeping pixels values as it is and changing only their positions makes a correlation coefficient better than altering the values. In terms of horizontal correlation, we discovered that using ILS-IP-2opt gives the best correlation of perturbed image with +0.1564614 while the (Gu, 2015) gives the toughest one. This means that our system perturbs the images while it maintains the utility of these last. In case of vertical correlation, ILS-2opt gives always the best answers with correlation coefficient equals to +0.1803171 while (Chen, 2015) always gives the worst correlation of perturbed image with the closest coefficient to the zero that equals to -0.0052000. Globally, we can state that our model yields a satisfactory solution in term of hiding the image significance and bearing on its utility.

De-Identification of Structured Data

The evaluation of de-identification of structured data was done using classification measures of data mining. As well as the other parts, this part of the system was compared with some conventional works that we have implemented on our data to get more credibility to the evaluation. For implementation of classification method, we have used the decision tree algorithm where the learning set was the original dataset and the test set was the perturbed dataset. We have applied these algorithms on real medical dataset collected from several hospitals in the USA. The aim of this algorithm is to classify the patients according to their medical classes (GENERAL, PAIN, NEUROLOGICAL, EQUILIBRIUM/PERCEPTION, SLEEP, EYES/VISION, COGNITIVE, EMOTIONAL, GASTROINTESTINAL, And UROGENITAL, SENSITIVITIES, SKIN, CARDIOVASCULAR (Heart), HAIR/NAILS, and OTHER). In other words, we have put our system in comparison with other algorithms within a

Table 8. Comparison of correlation coefficient of the two systems and the work presented in (Tong, 2009), (Chen, 2014), (Gu, 2014), and (Al-Najjar, 2011)

System		Correlation coefficient
Plain-Image	Horizontal	+0.9349843
	Vertical	+0.9378704
FWA-IP	Horizontal	+0.0323642
	Vertical	+0.1279019
ILS-IP-2opt	Horizontal	+0.1564614
	Vertical	+0.1803171
ILS-IP-3opt	Horizontal	+0.1368696
	Vertical	-0.0849279
(Tong, 2009)	Horizontal	+0.0171888
	Vertical	+0.0098527
(Chen, 2015)	Horizontal	-0.0061000
	Vertical	-0.0052000
(Gu, 2014)	Horizontal	+0.0139000
	Vertical	+0.0073000
(Al-Najjar, 2011)	Horizontal	+0.0186500
	Vertical	+0.0345000

multi-class's classification algorithm. The evaluation measures that we have used are precision, recall, F-measure with $\beta=1$ and $\beta=10$, and the entropy.

The results of table 9 above revealed that our system has given the best results in term of F-measure with $\beta=1$ with 97.4% and Entropy with 0.024%. meanwhile, the system presented in (McMurry, 2013) has given the best results in term of recall with 98% and the same with F-measure with $\beta=10$; but also this system has given the worst results where it has the highest entropy with 13.1% of lost data, and the lowest precision with 61.0%. We also noticed that the algorithm of (Szarvas, 2006) has revealed the best precision with 98.1%. Globally, we have noticed that our system has a satisfactory use in privacy preserving of structured data. The data mining results revealed that it can hide sensitive data while holding the essential information for utility in other statistical objectives.

Table 9. Results of de-identification of structured data

Approach	precision	recall	F-measure (1)	F-measure (10)	Entropy
Our approach	97.6%	97.2%	97.4%	97.2%	0.024%
(Fernandes, 2013)	95.1%	78.1%	85.8%	78.2%	02.1%
(McMurry, 2013)	61.0%	98.0%	75.0%	98.0%	13.1%
(Jaćimović, 2015) – Strict Criteria-	94.0%	94.0%	94.0%	93.1%	02.5%
(Jaćimović, 2015) – Relaxed Criteria -	97.0%	96.0%	97.0%	95.1%	01.3%
(Szarvas, 2006)	98.1%	96.7%	97.4%	95.8%	00.8%

Time of Answering

Finally, we admitted the results previously seen and analysed in term of time of replying in the table 10 bellow. We offered the both: time of responding when an authorised user has access to data, and when an unauthorised user wants to access to it. And for both types of data. We chose from each part the best result.

Table 10 represents the time of responding that our system takes in different cases, thus the type of treated data and the executed operations. For the measurement of time of answering we have executed our experiments a single node machine installed using vagrant on VIRTUALBOX that is installed on a machine which has the following configuration using apache spark:

- Processor: core i5-3230M CPU 2.56 Ghz
- Ram: 4096 Mo
- Swap: 1024 Mo

By analysing the results, we noticed that the verification of the signature is a required operation in which the system verifies the accessibility of a user to such data. Even so, we noted that the instance of accessing an unauthorised user of textual data takes a long time because of the wasted time in de-identification process. However, the de-identification of unstructured textual document of length 35971 words is the part that takes a long time with about 71 seconds, and that is due to the number of operations executed in for each antigen and detected anti-body, which means an exponential growth of number of operations for each iteration. For image data, we noticed that the de-identification of an image takes about 10 seconds which is a good time. Furthermore, our system has a satisfactory time of execution comparing to the used material. In real big data services that are applied using material with high capacity and multi node systems where we can execute several operations in parallel system, the time will be reduced to less than one second.

CONCLUSION

In this paper, we bring in suggestion a new model for privacy preserving by combining a mathematical access control model with bio-inspired based de-identification models for high security. Our approach is compounded on two major steps: the first one consists of defining users' identities, encrypt it, and generating access rights using the confidence interval and also generating digital signatures. The second step consists of verifying the access right corresponded to a user on such data using its digital signature. If the user has already the access to the data our system extends the original data in quasi-substantial time. Else our system consists of perturbing identifiable data either by using immune system for textual data or iterated local search for picture information. Our system presents a high level of security for the high complexity of used algorithms. Another advantage of our approach

Table 10. Responding time of our system in different cases

Case	Authorised user	Unauthorised user		
Time of responding	0.149 seconds	70.003 seconds	09.46 seconds	12.254 seconds
Data type	Any type (textual and image)	Unstructured Textual	Image	Structured data bases
Executed operations	Verification of signature	Verification of signature and de-identification	Verification of signature and de-identification	Verification of signature and de-identification

is that it preserves a maximum of data utility for de-identified one since it removes only identifier words and its variants in the case of textual data and does not change pixel values in case of image data. Finally, we concluded that our system presents a good tool for privacy over big data because it verifies almost the three major properties of big data:

- Variety of data, our system processes different data of different forms (textual, structured and image files).
- Velocity, instead of de-identified textual data, our system provides a rapid response to the users.
- Volume, because of the use of hierarchical identification, immune system, and ILS algorithms our system could be adapted to a parallel processing in which it can treat big volumes of data.
- Veracity, our system shows a high trustworthiness of the perturbed data by maintaining its utility.
- Value, even if this property has no relation with privacy concerns, but our system consists of hiding only sensitive information in a way that allows to anyone to transfer the generated data into a good value.

Thus, another disadvantage comes out in image de-identification part resides in the inefficiency of the approach and bi-colour images.

In the other hand, our system presents some impediments resides in the inability of predicting the appropriate values for both used threshold in the immune system and confidence level for access control. For that, and as perspectives, we will look for new methods and algorithms. Also, we will integrate image segmentation for geometric detection of the image's pixels as well as facial recognition in order to allow the perturbation of only descriptive pixels which theoretically seems as a good optimisation.

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